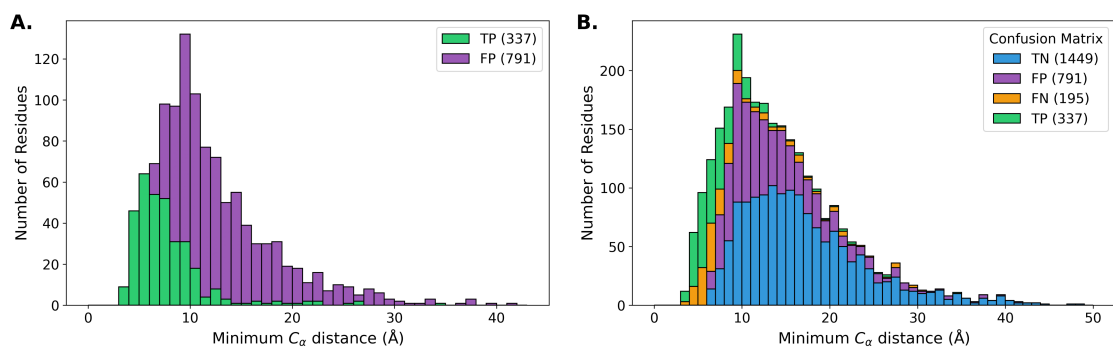


SI Table S2. Data for Hydrolase Receptors

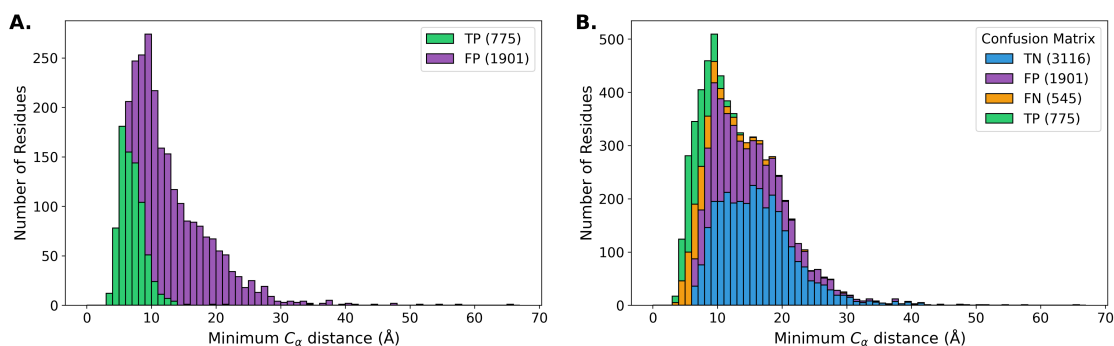
| apo_pdb | holo_pdb | apo_bmrh | holo_bmrh | F1   | MCC  | apo_pdb | holo_pdb | apo_bmrh | holo_bmrh | F1   | MCC   |
|---------|----------|----------|-----------|------|------|---------|----------|----------|-----------|------|-------|
| 2GS0    | 2LBM     | 15001    | 17569     | 0.60 | 0.46 | 7QCX    | 2JZI     | 51289    | 15624     | 0.43 | 0.22  |
|         | 6FDP     | 19757    | 34223     | 0.48 | 0.43 |         | 1D5G     | 34688    | 4516      | 0.27 | 0.22  |
|         | 2LOX     | 6225     | 18229     | 0.47 | 0.42 |         | 6OQJ     | 4276     | 30605     | 0.32 | 0.22  |
|         | 2PLD     | 5318     | 5310      | 0.46 | 0.40 |         | 2KWI     | 15524    | 15525     | 0.40 | 0.15  |
|         | 6OQJ     | 30599    | 30605     | 0.47 | 0.31 |         | 1VJ6     | 5131     | 6060      | 0.29 | 0.15  |
|         | 2RQG     | 4915     | 10019     | 0.43 | 0.30 |         | 1ONV     | 27288    | 5685      | 0.34 | 0.12  |
|         | 6GC3     | 17173    | 34260     | 0.29 | 0.28 | 2KQ3    | 2KHS     | 16585    | 15357     | 0.31 | 0.11  |
|         | 2LOZ     | 10318    | 17364     | 0.38 | 0.27 |         | 2LE8     | 16396    | 17702     | 0.21 | 0.10  |
|         | 1CF4     | 18251    | 4700      | 0.49 | 0.27 |         | 2HUG     | 6592     | 7241      | 0.35 | 0.09  |
|         | 2KC8     | 16066    | 16065     | 0.52 | 0.27 | 2L5U    | 2LTO     | 10276    | 18490     | 0.14 | 0.09  |
|         | 6R5G     | 28070    | 34384     | 0.39 | 0.25 |         | 2L75     | 17285    | 17344     | 0.44 | 0.02  |
|         | 2K7L     | 27288    | 15919     | 0.39 | 0.25 |         | 7NQC     | 51289    | 34608     | 0.34 | 0.01  |
| 2GS0    | 2KYL     | 15972    | 16967     | 0.41 | 0.23 | 2JNS    | 2RMK     | 5511     | 11010     | 0.18 | -0.02 |
|         | 2KWI     | 15230    | 15525     | 0.36 | 0.23 |         | 6BGG     | 15125    | 30367     | 0.14 | -0.08 |



SI Fig. S1. Confusion Matrix Histograms for Hydrolase Receptors

SI Table S3. Data for Transferase Receptors

| apo_pdb | holo_pdb | apo_bmr | holo_bmr | F1   | MCC  | apo_pdb | holo_pdb | apo_bmr | holo_bmr | F1   | MCC   |
|---------|----------|---------|----------|------|------|---------|----------|---------|----------|------|-------|
| 7JMY    | 2MRE     | 25071   | 25070    | 0.80 | 0.60 | 2L0I    | 2LVO     | 18610   | 18582    | 0.34 | 0.19  |
|         | 2N9P     | 25913   | 25914    | 0.72 | 0.58 |         | 2LAS     | 50128   | 25553    | 0.28 | 0.19  |
|         | 7JYN     | 30782   | 30790    | 0.53 | 0.43 |         | 5M9D     | 17044   | 34058    | 0.33 | 0.18  |
|         | 2N8T     | 25866   | 25865    | 0.64 | 0.42 |         | 2YS5     | 11095   | 11094    | 0.35 | 0.18  |
|         | 2K7A     | 11018   | 16809    | 0.54 | 0.42 | 6VED    | 2L3R     | 30703   | 17200    | 0.30 | 0.18  |
| 5AHT    | 2C52     | 15398   | 6874     | 0.56 | 0.41 |         | 5J8H     | 51289   | 30063    | 0.43 | 0.18  |
|         | 2PLD     | 5318    | 5310     | 0.46 | 0.40 |         | 2N83     | 25828   | 25833    | 0.36 | 0.17  |
|         | 2KPZ     | 25349   | 16574    | 0.55 | 0.40 |         | 5IAY     | 30703   | 30019    | 0.22 | 0.17  |
|         | 1W1F     | 1WA7    | 6261     | 0.51 | 0.38 |         | 2LXS     | 6095    | 18694    | 0.32 | 0.17  |
| 2LSY    | 2N1G     | 18455   | 25559    | 0.45 | 0.37 |         | 2MV7     | 18094   | 19516    | 0.38 | 0.15  |
|         | 2KWU     | 17769   | 16880    | 0.40 | 0.37 | 2MRE    | 7ZEY     | 16983   | 34727    | 0.44 | 0.14  |
| 2KM4    | 2L0I     | 16411   | 17044    | 0.36 | 0.36 |         | 2MRE     | 18610   | 25070    | 0.28 | 0.13  |
|         | 2KWV     | 17769   | 16885    | 0.39 | 0.34 |         | 5JYV     | 30092   | 30093    | 0.31 | 0.13  |
|         | 2LSJ     | 18431   | 18433    | 0.35 | 0.34 |         | 9C5E     | 18610   | 31177    | 0.32 | 0.13  |
|         | 2MC6     | 19428   | 19429    | 0.47 | 0.33 |         | 6QXZ     | 27250   | 27251    | 0.42 | 0.13  |
| 2LSY    | 2LSK     | 18455   | 18434    | 0.38 | 0.32 |         | 2H7D     | 15792   | 7150     | 0.24 | 0.11  |
|         | 2LVO     | 18581   | 18582    | 0.50 | 0.30 | 2M86    | 2M86     | 15945   | 19230    | 0.25 | 0.10  |
|         | 2KQ0     | 25349   | 16575    | 0.50 | 0.30 |         | 2N1A     | 6304    | 25553    | 0.10 | 0.10  |
|         | 2OJ2     | 4122    | 6581     | 0.37 | 0.28 |         | 2RSE     | 50325   | 11471    | 0.10 | 0.10  |
|         | 5VF0     | 17769   | 30276    | 0.31 | 0.27 |         | 2KA6     | 4789    | 16015    | 0.51 | 0.09  |
| 2JNS    | 5JYV     | 30091   | 30093    | 0.45 | 0.27 |         | 2LTO     | 10276   | 18490    | 0.14 | 0.09  |
|         | 1CF4     | 18251   | 4700     | 0.49 | 0.27 | 2K7A    | 2K7A     | 5461    | 16809    | 0.21 | 0.09  |
|         | 2NCZ     | 15125   | 26041    | 0.40 | 0.25 |         | 2KJE     | 4789    | 16318    | 0.45 | 0.07  |
|         | 2LGK     | 17813   | 17812    | 0.52 | 0.25 |         | 2MZD     | 34231   | 25484    | 0.35 | 0.06  |
|         | 2MOW     | 17173   | 19954    | 0.21 | 0.25 |         | 2RSY     | 7141    | 11508    | 0.30 | 0.05  |
| 2L0I    | 2MBB     | 15410   | 19394    | 0.43 | 0.24 |         | 2G35     | 15792   | 7061     | 0.17 | 0.04  |
|         | 5LVF     | 17044   | 34041    | 0.31 | 0.24 | 2MSR    | 2MSR     | 34179   | 25130    | 0.12 | 0.00  |
| 1U2N    | 1R8U     | 6268    | 5987     | 0.56 | 0.24 |         | 2RMK     | 5511    | 11010    | 0.18 | -0.02 |
|         | 1I5H     | 26698   | 4963     | 0.40 | 0.23 |         | 2LGG     | 17813   | 17808    | 0.37 | -0.03 |
|         | 2KYL     | 15972   | 16967    | 0.41 | 0.23 |         | 2I94     | 5332    | 7293     | 0.17 | -0.03 |
|         | 8SG2     | 27579   | 31080    | 0.44 | 0.23 |         | 2K8F     | 34231   | 15944    | 0.22 | -0.05 |
| 1U2N    | 1L8C     | 6268    | 5327     | 0.58 | 0.22 | 2JNS    | 2ND1     | 15125   | 26043    | 0.23 | -0.06 |
|         | 7ZEY     | 34724   | 34727    | 0.36 | 0.22 |         | 2LV6     | 51289   | 18556    | 0.35 | -0.07 |
|         | 2KA4     | 6268    | 16014    | 0.55 | 0.21 |         | 2N83     | 25883   | 25833    | 0.29 | -0.10 |
|         | 2MH0     | 34231   | 19610    | 0.33 | 0.21 |         | 9C5E     | 18642   | 31177    | 0.35 | -0.11 |
|         | 2MPS     | 6612    | 18876    | 0.34 | 0.20 |         | 2RSE     | 16931   | 11471    | 0.00 | -0.16 |
| 6U19    | 6U19     | 16620   | 27875    | 0.43 | 0.20 |         | 2JMF     | 6262    | 15016    | 0.13 | -0.17 |



SI Fig. S2. Confusion Matrix Histograms for Transferase Receptors